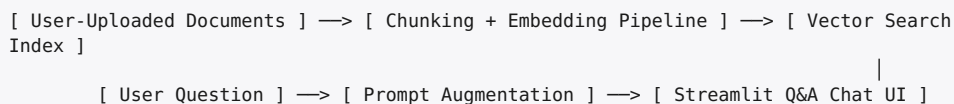


Custom RAG Agent — Project Notes

Grounded Question-Answering Over Your Own Documents



Main Goal

The Custom RAG Agent lets a user upload their own reference documents and then ask questions that get answered strictly from that material. Instead of relying on whatever the underlying model already knows, the agent treats the uploaded documents as the only source of truth.

How It Works

Uploaded documents are broken into chunks, converted into embeddings, and stored in a vector search index. When a question comes in, the agent searches that index for the pieces of the documents most relevant to the question, then hands those pieces to the model as the material it's allowed to work from. This retrieval step happens fresh for every question, so the answer is always built from whatever's actually relevant, not from a fixed summary of the whole document set.

Core Value

The system is built specifically to prevent hallucination. The prompt sent to the model is constrained to only use the retrieved material, with a clear fallback for when the documents don't contain enough to answer, and every claim in the response is tied back to a specific source passage. That combination — tight retrieval plus a tightly constrained prompt — is what keeps the answers grounded and verifiable instead of invented.

The Interface

The agent is built with LangChain and Python for the retrieval and generation pipeline, paired with a Streamlit chat interface where a user uploads documents, waits for them to be indexed, and then asks questions in a normal chat window, with answers streaming in and citations available to check against the original material.